

Adoption Friction in Multi-Stakeholder Construction Systems

In practice, adoption of high-performance construction systems is limited less by product quality and more by the need for predictable execution across: • designers • contractors • supervision engineers • safety authorities • and end users	Each actor optimizes for risk avoidance: • predictability • previous experience $\text{Risk} \times \text{Actors} = \text{Adoption Friction}$	Even when a system is: • technically superior • economically cheaper • operationally faster it still fails if any link in the chain resists deviation from established practice.
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The Real Barrier: Not Awareness, But Risk Avoidance

Even when stakeholders:

- understand technical benefits
- acknowledge efficiency gains
- recognize lifecycle cost advantages

They still default to known systems due to:

- perceived execution uncertainty
- fear of implementation errors
- dependency on prior experience
- lack of “routine familiarity”

Better systems often lose not in evaluation, but in execution comfort.

Multi-Stakeholder Dependency Problem

Unlike simple product sales, system adoption requires: synchronized behavioral acceptance across the entire project ecosystem If even one stakeholder (e.g. contractor or inspector) prefers familiar systems, the adoption is effectively blocked.	This is why adoption depends less on persuasion and more on: • repetition across projects • cross-role alignment • proven field reliability over time Adoption is not blocked by lack of value, but by lack of execution safety.
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Adoption is not decided at the point of sale, it is either enabled or blocked later in the execution chain, where no single stakeholder owns the outcome but each can prevent it.